

1 Fundamentals: Logic, sets, numbers and vectors

1.1 Real Numbers, Eulidean Spaces

Axioms of Addition:

- Axioms:

A1 (Associativity) $x + (y + z) = (x + y) + z \quad \forall x, y, z \in \mathbb{R}$

A2 (Neutral element) $x + 0 = x \quad \forall x \in \mathbb{R}$

A3 (Inverse element) $x + (-x) = 0 \quad \forall x \in \mathbb{R} \exists (-x) \in \mathbb{R}$

A4 (Commutativity) $x + y = y + x \quad \forall x, y \in \mathbb{R}$

Axioms of Multiplication:

- Axioms:

M1 (Associativity) $x \cdot (y \cdot z) = (x \cdot y) \cdot z \quad \forall x, y, z \in \mathbb{R}$

M2 (Neutral element) $x \cdot 1 = x \quad \forall x \in \mathbb{R}$

M3 (Inverse element) $x \cdot x^{-1} = 1 \quad \forall x \in \mathbb{R} \setminus \{0\} \exists x^{-1} \in \mathbb{R}$

M4 (Commutativity) $x \cdot z = z \cdot x \quad \forall x, z \in \mathbb{R}$

Distributivity (Compatibility of addition and mult.):

- Axiom:

D $x \cdot (y + z) = xy + xz = (y + z) \cdot x \quad \forall x, y, z \in \mathbb{R}$

Order Axioms:

- Axioms:

O1 (Reflexivity) $x \leq x \quad \forall x \in \mathbb{R}$

O2 (Transitivity) $(x \leq y \wedge y \leq z) \Rightarrow x \leq z$

O3 (Antisymmetry) $(x \leq y \wedge y \leq x) \Rightarrow x = y$

O4 (Totality) $\forall x, y \in \mathbb{R} : x \leq y \text{ or } y \leq x$

Compatibility (Consistency with addition and mult.):

- Axioms:

K1 $x \leq y \Rightarrow x + z \leq y + z \quad \forall x, y, z \in \mathbb{R}$

K2 $x \geq 0, y \geq 0 \Rightarrow xy \geq 0 \quad \forall x, y \in \mathbb{R}$

R 1.1.5:

• The set $\mathbb{Q}$ of rational numbers, equipped with addition, multiplication, and the order relation $\leq$, satisfies the above axioms.

V (Completeness axiom):

• The set $\mathbb{R}$ is **order complete**: For any two nonempty sets $A, B \subset \mathbb{R}$ such that

$$a \leq b \quad \text{for all } a \in A, b \in B,$$

there exists a number $c \in \mathbb{R}$ satisfying

$$a \leq c \leq b \quad \forall a \in A, b \in B.$$

C 1.1.6:

- The consequences of above axioms are:

1. Uniqueness of the additive and multiplicative inverse.

2. $0 \cdot x = 0 \quad \forall x \in \mathbb{R}$.

3. $(-1) \cdot x = -x \quad \forall x \in \mathbb{R}$, in particular $(-1)^2 = 1$.

4. $y \geq 0 \iff (-y) \leq 0$.

5. $y^2 \geq 0 \quad \forall y \in \mathbb{R}$, in particular $1 = 1 \cdot 1 \geq 0$.

6. If $x \leq y$ and $u \leq v$, then $x + u \leq y + v$.

7. If $0 \leq x \leq y$ and $0 \leq u$, then $xu \leq yu$.

T (Completeness of the real numbers):

- $\mathbb{R}$ is a commutative ordered field that is order-complete.

C 1.1.7 (Archimedean Principle):

- 1. For $x \in \mathbb{R}$ and $y > 0$ there exists $n \in \mathbb{N}$ such that

$$ny > x.$$

2. For every $\varepsilon > 0$ there exists $n \in \mathbb{N}$ such that

$$\frac{1}{n} < \varepsilon.$$

T 1.1.8:

- For every $t \geq 0$, $t \in \mathbb{R}$, the equation $x^2 = t$ has a solution in $\mathbb{R}$.

D (Bounded set - upper/lower bounded):

- A set $A \subset \mathbb{R}$ is called upper bounded if

$$\exists b \in \mathbb{R} \quad \forall a \in A : a \leq b.$$

Each such b is called an upper bound for A . (Similarly for lower bound.)

T (Supremum Infimum):

- 1. Maximum and minimum (if exists) are unique

2. Every non-empty subset bounded below (above) has a unique infimum (supremum).

3. $S = \sup(X) \iff$

$$(\forall x \in X : x \leq S) \wedge (\forall \varepsilon > 0 \exists x \in X : x > S - \varepsilon)$$

4. $I = \inf(X) \iff$

$$(\forall x \in X : x \geq I) \wedge (\forall \varepsilon > 0 \exists x \in X : x < I + \varepsilon)$$

D (Maximum):

- Maximum is:

$$\max\{x, y\} := \begin{cases} x, & \text{if } y \leq x, \\ y, & \text{if } x \leq y. \end{cases}$$

D (Minimum):

- Minimum is:

$$\min\{x, y\} := \begin{cases} y, & \text{if } y \leq x, \\ x, & \text{if } x \leq y. \end{cases}$$

T (Properties of the absolute value):

- Properties:

1. $|x| \geq 0$.

2. $|x + y| \leq |x| + |y|$ (triangle inequality).

3. $|x + y| \geq ||x| - |y||$ (inv. triangle inequality).

T (Young's inequality):

- For every $\varepsilon > 0$ and all $x, y \in \mathbb{R}$ it holds:

$$2|xy| \leq \varepsilon x^2 + \frac{1}{\varepsilon} y^2.$$

1.2 Vectors and Vector Spaces

D (inner product):

- For $x, y \in \mathbb{R}^n$ the standard inner product is

$$x \cdot y = \langle x, y \rangle = \sum_{i=1}^n x_i y_i$$

D (Euclidean norm):

- The Euclidean norm is

$$\|x\| = \sqrt{\sum_{i=1}^n x_i^2}$$

D (Euclidean distance):

- The Euclidean distance is

$$d(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

D (Triangle inequality):

- For all $x, y, z \in \mathbb{R}^n$:

$$\|x - z\| \leq \|x - y\| + \|y - z\|$$

1.3 Complex Numbers

D (Complex number):

- A complex number is

$$z = a + ib, \quad a, b \in \mathbb{R}$$

D (set of complex numbers):

- The set of complex numbers is

$$\mathbb{C} = \{a + ib \mid a, b \in \mathbb{R}\}$$

D (Terminology of complex numbers):

- Terminology
 - $a = \operatorname{Re}(z)$ — real part
 - $b = \operatorname{Im}(z)$ — imaginary part
 - $i^2 = -1$
 - If $a = 0$, the number is purely imaginary

D (Complex numbers Addition and Subtraction):

- Let $z_1 = x_1 + iy_1$, $z_2 = x_2 + iy_2$.

$$z_1 + z_2 = (x_1 + x_2) + i(y_1 + y_2)$$

$$z_1 - z_2 = (x_1 - x_2) + i(y_1 - y_2)$$

D (Complex numbers Multiplication):

- Let $z_1 = x_1 + iy_1$, $z_2 = x_2 + iy_2$.

$$z_1 z_2 = x_1 x_2 - y_1 y_2 + i(x_1 y_2 + y_1 x_2)$$

D (Complex conjugate):

- For a complex number $z = x + iy$, the complex conjugate of z is given by

$$\bar{z} = \overline{x + iy} = x - iy.$$

D (Complex numbers Division):

- To divide numbers we use conjugate:

$$\frac{z}{w} = \frac{z\bar{w}}{w\bar{w}} = \frac{z\bar{w}}{|w|^2}$$

D (Properties of the Conjugate):

- Properties are:

- $z\bar{z} = |z|^2$
- $z + \bar{z} = 2\operatorname{Re}(z)$, $z - \bar{z} = 2i\operatorname{Im}(z)$
- $\bar{\bar{z}} = z$
- $\overline{z \pm w} = \bar{z} \pm \bar{w}$
- $\overline{z\bar{w}} = \bar{z}w$
- $z = \bar{z}$, if $z \in \mathbb{R}$
- $\bar{z} = -z$, z is purely imaginary

D (Modulus):

- The magnitude $|z|$ of a complex number $z = a + ib$ is the length of the corresponding vector.

$$|z| = \sqrt{a^2 + b^2}$$

The distance between two complex numbers z_1 and z_2 is

$$d(z_1, z_2) = |z_2 - z_1|$$

Triangle inequality works also for complex numbers:

$$|z + w| \leq |z| + |w|$$

D (Euler's Formula):

- The exponential function can be represented by a series expansion as

$$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$$

If we substitute $x = it$ into this formula, we obtain Euler's formula.

$$e^{it} = \cos t + i \sin t$$

We can write $\sin(t)$ and $\cos(t)$ using complex exponential functions.

$$\cos(t) = \frac{e^{it} + e^{-it}}{2}$$

$$\sin(t) = \frac{e^{it} - e^{-it}}{2i}$$

D (Exponential with Complex Argument):

- We can also allow complex arguments for the exponential function. Then the following rules apply:

- $e^{z+w} = e^z e^w$
- $e^{a+ib} = e^a (\cos b + i \sin b)$
- $e^{z+i2\pi} = e^z$

D (Polar Form):

- A complex number can also be described by specifying the distance from the origin and by specifying a suitable angle (**polar form**):
 $z = re^{i\varphi}$.

- $r = |z| \geq 0$ - distance
- $\varphi \in (-\pi, \pi]$ - polar angle
- $\varphi = \arg(z) = \arccos(z)$

D (Polar to Cartesian conversion):

- Polar to Cartesian conversion is:

- $x = r \cos \varphi$
- $y = r \sin \varphi$

D (Cartesian to Polar conversion):

- Cartesian to Polar conversion is:

- $r = \sqrt{x^2 + y^2}$
- $\varphi = \arctan \frac{y}{x}$

If $x = 0$:

$$\arg(iy) = \begin{cases} \frac{\pi}{2}, & y > 0 \\ -\frac{\pi}{2}, & y < 0 \end{cases}$$

D (Powers of Complex Numbers):

- The following rules must be observed:

- $z_1 z_2 = r_1 r_2 e^{i(\varphi_1 + \varphi_2)}$
- $\frac{z_1}{z_2} = \frac{r_1}{r_2} e^{i(\varphi_1 - \varphi_2)}$
- $z^n = r^n e^{in\varphi}$

D (Roots from complex numbers):

- For $n \in \mathbb{N}$, the solutions to the equation $z^n = re^{i\varphi}$ are

$$z_k = r^{1/n} e^{i(\varphi/n + 2\pi k/n)}, \quad k = 0, \dots, n-1$$

D (Quadratic Equations):

- For the quadratic equation

$$az^2 + bz + c = 0,$$

in complex numbers the solution is still

$$z_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

D (Fundamental Theorem of Algebra):

- Each polynomial

$$p(z) = \sum_{k=0}^n a_k z^k, \quad a_k \in \mathbb{C}$$

can be factored into linear factors, i.e. written as

$$p(z) = a_n (z - z_1)(z - z_2) \dots (z - z_n)$$

The numbers z_k are therefore precisely the zero points of $p(z)$ (with **multiplicity**).

C (non-real roots appear in conjugate pairs):

- Non-real roots of a polynomial with **real** coefficients occur in conjugate pairs $z, \bar{z}$.

2 Sequences

D (Sequence):

• A sequence $(a_n)_{n \in \mathbb{N}_0}$ (also written $(a_n)_{n \geq 0}$ or $(a_n)_{n=0}^\infty$) is an infinite list of numbers, where each entry is called a *term* of the sequence. A sequence can also be viewed as a function

$$f : \mathbb{N}_0 \rightarrow \mathbb{R} \quad (\text{or } f : \mathbb{N} \rightarrow \mathbb{R}),$$

whose values are $f(n) = a_n$.

D (Ways to Describe a Sequence):

• One way to describe a sequence is by giving an explicit formula for each term:

$$a_n = f(n).$$

Another possibility is to define the sequence recursively, constructing each new term from previous ones:

$$a_n = g(a_{n-1}, a_{n-2}, \dots).$$

The functions f and g are called the *rule of formation* of the sequence.

D (Convergence/Divergence):

• A sequence $(a_n)_{n \in \mathbb{N}_0} \subset \mathbb{R}$ is called *convergent* if there exists a number $L \in \mathbb{R}$ such that

$$\forall \varepsilon > 0 \exists N > 0 \text{ such that } \forall n > N : |a_n - L| < \varepsilon.$$

The number L is called the *limit* of the sequence, and we write

$$\lim_{n \rightarrow \infty} a_n = L.$$

If no such L exists, the sequence is called *divergent*.

T (Uniqueness of the Limit):

• A convergent sequence has exactly one unique limit.

D (Divergence to infinity):

• Let $(a_n)_{n \in \mathbb{N}_0}$ be a sequence.

• If

$$\forall M > 0 \exists N > 0 \text{ such that } \forall n > N : a_n > M,$$

then the sequence diverges to infinity and we write

$$\lim_{n \rightarrow \infty} a_n = \infty.$$

• If

$$\forall M > 0 \exists N > 0 \text{ such that } \forall n > N : a_n < -M,$$

then the sequence diverges to minus infinity and we write

$$\lim_{n \rightarrow \infty} a_n = -\infty.$$

D (Subsequence):

• Let $(a_n)_{n \in \mathbb{N}_0}$ be a sequence in $\mathbb{R}$. A *subsequence* is a sequence of the form

$$(a_{n_k})_{k \in \mathbb{N}_0},$$

where $(n_k)_{k \in \mathbb{N}_0}$ is a sequence of non-negative integers satisfying

$$n_{k+1} > n_k \quad \text{for all } k \in \mathbb{N}_0.$$

L (Limit of a sequence and its subsequence):

• Let $(a_n)_{n \in \mathbb{N}_0}$ be a convergent sequence with limit $L \in \mathbb{R}$. Then every subsequence $(a_{n_k})_{k \in \mathbb{N}_0}$ also converges to the same limit L .

D (Accumulation Point):

• Let $(a_n)_{n \in \mathbb{N}_0}$ be a sequence in $\mathbb{R}$. A number $A \in \mathbb{R}$ is called an accumulation point of the sequence if

$$\forall \varepsilon > 0 \forall N \in \mathbb{N}_0 \exists n \geq N \text{ such that } |a_n - A| < \varepsilon.$$

Intuitively, every interval $(A - \varepsilon, A + \varepsilon)$ contains terms a_n for infinitely many indices n .

T (Characterization of Accumulation Points):

• Let $(a_n)_{n \in \mathbb{N}_0}$ be a sequence in $\mathbb{R}$. A number $A \in \mathbb{R}$ is an accumulation point of the sequence if and only if there exists a convergent subsequence $(a_{n_k})_{k \in \mathbb{N}_0}$ such that

$$\lim_{k \rightarrow \infty} a_{n_k} = A.$$

C (Accumulation Point of a Convergent Sequence):

• Every convergent sequence $(a_n)_{n \in \mathbb{N}_0}$ in $\mathbb{R}$ has exactly one accumulation point, which coincides with its limit.

C (Infinite Occurrence Near an Accumulation Point):

• Let A be an accumulation point of the sequence $(a_n)_{n \in \mathbb{N}_0}$. Then for every $\varepsilon > 0$ there exist infinitely many terms of the sequence lying in the interval $(A - \varepsilon, A + \varepsilon)$.

T (Algebra of Limits):

• Assume

$$\lim_{n \rightarrow \infty} a_n = K, \quad \lim_{n \rightarrow \infty} b_n = L$$

with $K, L \neq \pm\infty$ and let C be a constant. Then

$$\lim_{n \rightarrow \infty} (a_n + b_n) = K + L$$

$$\lim_{n \rightarrow \infty} (a_n - b_n) = K - L$$

$$\lim_{n \rightarrow \infty} (a_n b_n) = KL$$

$$\lim_{n \rightarrow \infty} (C a_n) = CK$$

If $L \neq 0$ and $b_n \neq 0$ then

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \frac{K}{L}.$$

If $K < L$ then there exists N such that $a_n < b_n$ for all $n \geq N$. If there exists N such that $a_n \leq b_n$ for all $n \geq N$, then $K \leq L$.

T (Important Limits):

•

$$\lim_{n \rightarrow \infty} \frac{\log n}{n} = 0 = \lim_{n \rightarrow \infty} \frac{\ln n}{n}$$

$$\lim_{n \rightarrow \infty} n^{1/n} = 1$$

$$\lim_{n \rightarrow \infty} x^{1/n} = 1 \quad (x > 0)$$

$$\forall x \in \mathbb{R} : \lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n = e^x$$

$$\forall x \in \mathbb{R} : \lim_{n \rightarrow \infty} \frac{x^n}{n!} = 0$$

T (Sandwich Theorem):

• Let $(a_n)_{n \in \mathbb{N}_0}$ and $(c_n)_{n \in \mathbb{N}_0}$ be convergent sequences with

$$\lim_{n \rightarrow \infty} a_n = L, \quad \lim_{n \rightarrow \infty} c_n = L$$

and suppose there exists N_0 such that

$$a_n \leq b_n \leq c_n \quad \text{for all } n \geq N_0.$$

Then $(b_n)_{n \in \mathbb{N}_0}$ is also convergent and

$$\lim_{n \rightarrow \infty} b_n = L.$$

D (Bounded Sequence):

• A sequence $(a_n)_{n \in \mathbb{N}_0}$ is called bounded below or bounded above if the set

$$M = \{a_n \mid n \in \mathbb{N}_0\}$$

has a lower bound or an upper bound respectively. A sequence that is bounded both above and below is called a bounded sequence.

D (Monotone Sequence):

• A sequence $(a_n)_{n \in \mathbb{N}_0}$ is called monotone decreasing if

$$m > n \Rightarrow a_m \leq a_n$$

and monotone increasing if

$$m > n \Rightarrow a_m \geq a_n.$$

It is called strictly monotone decreasing if

$$m > n \Rightarrow a_m < a_n$$

and strictly monotone increasing if

$$m > n \Rightarrow a_m > a_n.$$

T (Monotone Convergence Theorem):

- Every bounded and monotone sequence converges.

L (Convergent sequence = bounded):

- Every convergent sequence is bounded.

T Monotone Convergence:

- Let $(a_n)_{n \in \mathbb{N}_0}$ be a sequence of real numbers.

(i) A monotone sequence converges if and only if it is bounded.

(ii) If (a_n) is monotonically increasing, then

$$\lim_{n \rightarrow \infty} a_n = \sup\{a_n \mid n \in \mathbb{N}_0\}.$$

(iii) If (a_n) is monotonically decreasing, then

$$\lim_{n \rightarrow \infty} a_n = \inf\{a_n \mid n \in \mathbb{N}_0\}.$$

D Limit Superior and Limit Inferior:

- Let $(a_n)_{n \in \mathbb{N}_0}$ be a sequence.

- *limit superior* of the sequence

$$\limsup_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \sup\{a_k \mid k \geq n\}$$

- *limit inferior* of the sequence.

$$\liminf_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \inf\{a_k \mid k \geq n\}$$

L Properties of limsup and liminf:

- Let $(a_n)_{n \in \mathbb{N}_0}$ be a sequence. Then:

(i) $\limsup_{n \rightarrow \infty} a_n = \inf_{n \in \mathbb{N}} (\sup\{a_k \mid k \geq n\})$

(ii) $\liminf_{n \rightarrow \infty} a_n = \sup_{n \in \mathbb{N}} (\inf\{a_k \mid k \geq n\})$

(iii) If (a_n) is convergent, then

$$\lim_{n \rightarrow \infty} a_n = \limsup_{n \rightarrow \infty} a_n = \liminf_{n \rightarrow \infty} a_n.$$

(iv) A bounded sequence (a_n) converges if and only if

$$\lim_{n \rightarrow \infty} a_n = \limsup_{n \rightarrow \infty} a_n = \liminf_{n \rightarrow \infty} a_n.$$

T Characterization via limsup:

- Let $(a_n)_{n \in \mathbb{N}_0}$ be a bounded sequence and define

$$A = \limsup_{n \rightarrow \infty} a_n.$$

Then:

(i) A is an accumulation point of the sequence.

(ii) For every $\varepsilon > 0$, there are only finitely many indices n such that

$$a_n > A + \varepsilon.$$

(iii) For every $\varepsilon > 0$, there are infinitely many indices n such that

$$A - \varepsilon < a_n < A + \varepsilon.$$

An analogous statement holds for the limit inferior.

C Subsequence of a Bounded Sequence:

- Every bounded sequence of real numbers has at least one accumulation point and admits a convergent subsequence.

D Cauchy Sequence:

- A sequence $(a_n)_{n \in \mathbb{N}_0} \subset \mathbb{R}$ is called a *Cauchy sequence* if

$$\forall \varepsilon > 0 \exists N \in \mathbb{N} \text{ such that } \forall m, n \geq N : |a_n - a_m| < \varepsilon.$$

L Cauchy Sequences are Bounded:

- Every Cauchy sequence is bounded.

T Cauchy Criterion:

- A sequence of real numbers converges if and only if it is a Cauchy sequence.

3 Series

D Series:

- A (formal) expression of the form

$$a_0 + a_1 + a_2 + \cdots = \sum_{n=0}^{\infty} a_n$$

is called a *series*. The terms a_n are called elements or summands. If the sequence of partial sums

$$s_n := \sum_{k=0}^n a_k$$

converges to a limit $L \in \mathbb{R}$, then the series is said to *converge*, and we write

$$\sum_{n=0}^{\infty} a_n = L.$$

Otherwise, the series is called *divergent*.

T Necessary Condition for Convergence:

- If the series $\sum_{n=0}^{\infty} a_n$ converges, then necessarily

$$\lim_{n \rightarrow \infty} a_n = 0.$$

L Linearity of Series:

- Let $\sum_{n=0}^{\infty} a_n$ and $\sum_{n=0}^{\infty} b_n$ be convergent series. Then:

$$\begin{aligned} \sum_{n=0}^{\infty} (a_n + b_n) &= \sum_{n=0}^{\infty} a_n + \sum_{n=0}^{\infty} b_n, \\ \sum_{n=0}^{\infty} C a_n &= C \sum_{n=0}^{\infty} a_n \quad (C \in \mathbb{R}). \end{aligned}$$

L Tail of a Series:

- Let $\sum_{n=0}^{\infty} a_n$ be a series. For any $N \in \mathbb{N}_0$:

$$\sum_{n=0}^{\infty} a_n \text{ converges} \iff \sum_{n=N}^{\infty} a_n \text{ converges,}$$

and in that case

$$\sum_{n=0}^{\infty} a_n = \sum_{n=0}^{N-1} a_n + \sum_{n=N}^{\infty} a_n.$$

T Series with Non-Negative Terms:

- Let $a_n \geq 0$ for all n . Then the sequence of partial sums

$$s_n = \sum_{k=0}^n a_k$$

is monotonically increasing. If (s_n) is bounded, then the series $\sum_{n=0}^{\infty} a_n$ converges; otherwise, it diverges.

T Comparison Test:

- Let $\sum a_n, \sum b_n, \sum c_n$ be series with non-negative terms and suppose that for some N :

$$c_n \leq b_n \leq a_n \quad \text{for all } n \geq N.$$

Then:

- If $\sum a_n$ converges, then $\sum b_n$ converges (majorant test).
- If $\sum c_n$ diverges, then $\sum b_n$ diverges (minorant test).

D Absolute and Conditional Convergence:

- A series $\sum a_n$ is called *absolutely convergent* if $\sum |a_n|$ converges.
- It is called *conditionally convergent* if $\sum a_n$ converges but $\sum |a_n|$ does not.

T Riemann Rearrangement Theorem:

- Let $\sum a_n$ be a conditionally convergent series of real numbers, and let $L \in \mathbb{R}$. Then there exists a bijection $\varphi : \mathbb{N}_0 \rightarrow \mathbb{N}_0$ such that

$$\sum_{n=0}^{\infty} a_{\varphi(n)} = L.$$

T Rearrangement of Absolutely Convergent Series:

- Let $\sum a_n$ be absolutely convergent and let $\varphi : \mathbb{N}_0 \rightarrow \mathbb{N}_0$ be a bijection. Then

$$\sum_{n=0}^{\infty} a_{\varphi(n)}$$

converges absolutely and

$$\sum_{n=0}^{\infty} a_n = \sum_{n=0}^{\infty} a_{\varphi(n)}.$$

T Leibniz Criterion:

- Let (a_n) be a monotonically decreasing sequence of non-negative real numbers with

$$\lim_{n \rightarrow \infty} a_n = 0.$$

Then the alternating series

$$\sum_{n=0}^{\infty} (-1)^n a_n$$

converges. Moreover, for all $m \in \mathbb{N}_0$:

$$\sum_{n=0}^{2m+1} (-1)^n a_n \leq \sum_{n=0}^{\infty} (-1)^n a_n \leq \sum_{n=0}^{2m} (-1)^n a_n.$$

T Cauchy Criterion for Series:

- The series $\sum_{n=0}^{\infty} a_n$ converges if and only if for every $\varepsilon > 0$ there exists N such that for all $n \geq m \geq N$:

$$\left| \sum_{k=m+1}^n a_k \right| \leq \varepsilon.$$

T Absolute Convergence Implies Convergence:

- Every absolutely convergent series converges, and

$$\left| \sum_{n=0}^{\infty} a_n \right| \leq \sum_{n=0}^{\infty} |a_n|.$$

T Root Test:

- Let (a_n) be a sequence and define

$$\rho = \limsup_{n \rightarrow \infty} |a_n|^{1/n}.$$

Then:

- If $\rho < 1$, the series $\sum a_n$ converges absolutely.
- If $\rho > 1$, the series $\sum a_n$ diverges.

T Ratio Test:

- Let (a_n) with $a_n \neq 0$ and define

$$\rho = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|.$$

Then:

- If $\rho < 1$, the series $\sum a_n$ converges absolutely.
- If $\rho > 1$, the series $\sum a_n$ diverges.

T Cauchy Product:

- Let $\sum a_n$ and $\sum b_n$ be absolutely convergent. Then

$$\left(\sum_{n=0}^{\infty} a_n \right) \left(\sum_{n=0}^{\infty} b_n \right) = \sum_{n=0}^{\infty} \left(\sum_{k=0}^n a_{n-k} b_k \right),$$

and the series on the right converges absolutely.

D Power Series:

- A series of the form

$$\sum_{k=0}^{\infty} c_k (x - a)^k$$

is called a *power series* centered at a with coefficients c_k .

D Convergence of a Power Series:

- For fixed x , if the sequence of partial sums

$$s_n(x) = \sum_{k=0}^n c_k (x - a)^k$$

converges, then the power series is said to converge at x .

T Radius of Convergence:

- Every power series has a radius of convergence R such that:

- The series converges for $|x - a| < R$,
- The series diverges for $|x - a| > R$.

The interval $(a - R, a + R)$ is called the interval of convergence. The radius R is given by:

$$\rho = \limsup_{n \rightarrow \infty} |c_n|^{1/n}.$$

Then

$$R = \begin{cases} 0 & \text{if } \rho = \infty, \\ \frac{1}{\rho} & \text{if } 0 < \rho < \infty, \\ \infty & \text{if } \rho = 0. \end{cases}$$

4 Functions

4.1 Function basics

D Function:

• A function (mapping) is a rule that assigns to each admissible input exactly one output in the target set. The set of all possible inputs is called the *domain*, denoted by $\text{domain}(f)$ or D . The set of all actually attained outputs is called the *image*, denoted by $\text{image}(f)$ or W .

$$f : \text{domain}(f) \rightarrow \text{range}(f), \quad x \mapsto f(x).$$

The input is called the *independent variable*, the output the *dependent variable*.

D Restriction of a Function:

• Let $f : D(f) \rightarrow \mathbb{R}$ and let $D_0 \subseteq D(f)$. The restriction of f to D_0 is the function

$$f|_{D_0} : D_0 \rightarrow \mathbb{R}, \quad f|_{D_0}(x) = f(x).$$

Note that f and $f|_{D_0}$ are, in general, different functions.

D Bounded Functions:

• Let $f : D(f) \rightarrow \mathbb{R}$.

• f is *bounded above* if there exists $M \in \mathbb{R}$ such that

$$f(x) \leq M \quad \forall x \in D(f).$$

• f is *bounded below* if there exists $M \in \mathbb{R}$ such that

$$f(x) \geq M \quad \forall x \in D(f).$$

• f is *bounded* if there exists $M \in \mathbb{R}$ such that

$$|f(x)| \leq M \quad \forall x \in D(f).$$

D Injective, Surjective, Bijective:

• Let $f : X \rightarrow Y$.

• f is *injective* if

$$f(x_1) = f(x_2) \Rightarrow x_1 = x_2.$$

• f is *surjective* if

$$\forall y \in Y \exists x \in X : f(x) = y.$$

• f is *bijective* if it is both injective and surjective.

D Composition:

• Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$. The composition is defined by

$$g \circ f : X \rightarrow Z, \quad (g \circ f)(x) = g(f(x)).$$

D Identity and Inverse:

• For a set X , the identity function is defined by

$$\text{id}_X(x) = x \quad \forall x \in X.$$

If $f : X \rightarrow Y$ is bijective, then there exists a unique function

$$f^{-1} : Y \rightarrow X$$

such that

$$f^{-1} \circ f = \text{id}_X, \quad f \circ f^{-1} = \text{id}_Y.$$

This function is called the *inverse* of f .

D Monotonicity:

• Let $f : D(f) \subseteq \mathbb{R} \rightarrow \mathbb{R}$.

• f is *monotonically increasing* if

$$x_1 < x_2 \Rightarrow f(x_1) \leq f(x_2).$$

• f is *strictly increasing* if

$$x_1 < x_2 \Rightarrow f(x_1) < f(x_2).$$

• f is *monotonically decreasing* if

$$x_1 < x_2 \Rightarrow f(x_1) \geq f(x_2).$$

• f is *strictly decreasing* if

$$x_1 < x_2 \Rightarrow f(x_1) > f(x_2).$$

T Strict Monotonicity Implies Injectivity:

• Every strictly monotone function is injective.

D Even and Odd Functions:

• Let $f : \mathbb{R} \rightarrow \mathbb{R}$.

• f is *even* if

$$f(-x) = f(x).$$

• f is *odd* if

$$f(-x) = -f(x).$$

D Convexity and Concavity (Geometric):

• The graph of a function is called

• *convex* if it bends to the left when moving from left to right,

• *concave* if it bends to the right when moving from left to right.

4.2 Limits of Functions

D Limit of a Function:

• Let $f : D(f) \rightarrow \mathbb{R}$ and $x_0 \in \mathbb{R}$. A number $L \in \mathbb{R}$ is called the limit of $f(x)$ at x_0 if

$$\forall \varepsilon > 0 \exists \delta > 0 \text{ such that } x \in D(f), |x - x_0| < \delta \Rightarrow |f(x) - L| < \varepsilon.$$

The limit is uniquely determined.

D One-Sided Limits:

• One-Sided Limits

• Right-hand limit:

$$\lim_{x \rightarrow x_0^+} f(x) = L$$

• Left-hand limit:

$$\lim_{x \rightarrow x_0^-} f(x) = L$$

D Limits at Infinity:

• Limits at Infinity

$$\lim_{x \rightarrow \infty} f(x) = L$$

$$\lim_{x \rightarrow -\infty} f(x) = L$$

D Infinite Limits:

• Infinite Limits

$$\lim_{x \rightarrow c} f(x) = \infty$$

$$\lim_{x \rightarrow c} f(x) = -\infty$$

T Limit Laws:

• Let

$$\lim_{x \rightarrow c} f(x) = K, \quad \lim_{x \rightarrow c} g(x) = L, \quad K, L \neq \pm\infty.$$

Then:

$$\lim_{x \rightarrow c} (f + g) = K + L$$

$$\lim_{x \rightarrow c} (f - g) = K - L$$

$$\lim_{x \rightarrow c} (fg) = KL$$

$$\lim_{x \rightarrow c} (Af) = AK$$

$$\text{• If } L \neq 0: \lim_{x \rightarrow c} \frac{f}{g} = \frac{K}{L}$$

4.3 Continuity

D Continuity:

- Let $f : D \rightarrow \mathbb{R}$ and $x_0 \in D$. The function f is called *continuous at x_0* if

$$\forall \varepsilon > 0 \exists \delta > 0 \text{ such that } |x - x_0| < \delta \Rightarrow |f(x) - f(x_0)| < \varepsilon.$$

The function is called *continuous* if it is continuous at every point in its domain.

D One-Sided Continuity:

- Let $f : D(f) \rightarrow \mathbb{R}$ and $x_0 \in D(f)$.
 - f is *right-continuous* at x_0 if

$$\lim_{x \rightarrow x_0^+} f(x) = f(x_0).$$

- f is *left-continuous* at x_0 if

$$\lim_{x \rightarrow x_0^-} f(x) = f(x_0).$$

T Sequential Characterization of Continuity:

- Let $f : D \subseteq \mathbb{R} \rightarrow \mathbb{R}$ and $x_0 \in D$. Then f is continuous at x_0 if and only if for every sequence (x_n) with $x_n \rightarrow x_0$ we have

$$f(x_n) \rightarrow f(x_0).$$

T Algebra of Continuous Functions:

- Let $f, g : D \subseteq \mathbb{R} \rightarrow \mathbb{R}$ be continuous. Then:

- $f + g$ is continuous
- $f - g$ is continuous
- cf is continuous for any constant $c \in \mathbb{R}$
- $f \cdot g$ is continuous
- $\frac{f}{g}$ is continuous if $g \neq 0$

T Continuity of Composition:

- Let $f : I \rightarrow \text{image}(f)$ and $g : \text{image}(f) \rightarrow \mathbb{R}$ be continuous. Then the composition

$$g \circ f$$

is continuous and

$$\lim_{x \rightarrow x_0} g(f(x)) = g\left(\lim_{x \rightarrow x_0} f(x)\right).$$

T Continuity of the Inverse:

- Let I be an interval and let $f : I \rightarrow J$ be continuous and bijective. Then J is also an interval and the inverse function

$$f^{-1} : J \rightarrow I$$

is continuous.

T Intermediate Value Theorem:

- Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous and let c be a value between $f(a)$ and $f(b)$. Then there exists $x \in [a, b]$ such that

$$f(x) = c.$$

C Zero of a Continuous Function (Bolzano theorem):

- Let $f : I \subseteq \mathbb{R} \rightarrow \mathbb{R}$ be continuous and suppose

$$f(a) < 0, \quad f(b) > 0.$$

Then there exists $x \in (a, b)$ such that

$$f(x) = 0.$$

D Compact Interval:

- A bounded and closed interval $[a, b]$ is called *compact*.

L Bolzano–Weierstrass for Intervals:

- Let (x_n) be a sequence in a compact interval $[a, b]$. Then there exists a subsequence (x_{n_k}) such that

$$x_{n_k} \rightarrow x_0 \quad \text{with } x_0 \in [a, b].$$

D Local Extrema:

- Let $f : D \subseteq \mathbb{R} \rightarrow \mathbb{R}$.

- x_0 is a *local maximum* if

$$f(x_0) \geq f(x) \quad \forall x \in D.$$

- x_0 is a *local minimum* if

$$f(x_0) \leq f(x) \quad \forall x \in D.$$

- Local maxima and minima are called *local extrema*.

T Extreme Value Theorem:

- Let $f : I \subseteq \mathbb{R} \rightarrow \mathbb{R}$ be continuous and let I be compact. Then f is bounded and attains its maximum and minimum on I , i.e., there exist $x_{\min}, x_{\max} \in I$ such that

$$f(x_{\min}) \leq f(x) \leq f(x_{\max}) \quad \forall x \in I.$$

5 Derivatives

5.1 Differentiation

D Derivative:

• Let $f : D \rightarrow \mathbb{R}$ and assume D has no isolated points. The *difference quotient* is

$$\frac{f(a+h) - f(a)}{h}.$$

If the limit exists, the *derivative* of f at a is defined by

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} = \lim_{b \rightarrow a} \frac{f(b) - f(a)}{b - a}.$$

Interpretation:

- instantaneous rate of change
- slope of the tangent
- sign indicates increase/decrease

D Derivative Function:

• Given $f : D \rightarrow \mathbb{R}$, the *derivative function* is

$$x \mapsto f'(x).$$

D Differentiability:

• A function f is *differentiable at x* if $f'(x)$ exists. If this holds for all x in the domain, f is called differentiable.

L Differentiability Implies Continuity:

• If f is differentiable at x_0 , then f is continuous at x_0 .

D One-Sided Derivatives:

•

• Right derivative:

$$\lim_{h \rightarrow 0^+} \frac{f(a+h) - f(a)}{h}$$

• Left derivative:

$$\lim_{h \rightarrow 0^-} \frac{f(a+h) - f(a)}{h}$$

D Higher Derivatives:

• Define iteratively:

$$f^{(0)} = f, \quad f^{(1)} = f', \quad f^{(2)} = f'', \quad f^{(n+1)} = (f^{(n)})'.$$

If $f^{(n)}$ exists, f is n -times differentiable. If all derivatives up to order n are continuous, then $f \in C^n(D)$. Define

$$C^\infty(D) := \bigcap_{n=0}^{\infty} C^n(D),$$

called *smooth functions*.

T Basic Derivatives:

•

• Power:

$$(ax^p)' = apx^{p-1}$$

• Exponential:

$$(a^x)' = \ln(a) a^x$$

• Trigonometric:

$$(\sin x)' = \cos x, \quad (\cos x)' = -\sin x, \quad (\tan x)' = \frac{1}{\cos^2 x}$$

• Hyperbolic:

$$(\sinh x)' = \cosh x, \quad (\cosh x)' = \sinh x$$

• Inverse:

$$(\ln x)' = \frac{1}{x}, \quad (\arctan x)' = \frac{1}{1+x^2}$$

T Sum and Product Rule:

• For differentiable f, g :

$$(f+g)' = f' + g'$$

$$(fg)' = f'g + fg'$$

T Chain Rule:

• If f is differentiable at x_0 and g at $f(x_0)$, then

$$(g \circ f)'(x_0) = g'(f(x_0)) \cdot f'(x_0).$$

T Quotient Rule:

• If $g(x_0) \neq 0$, then

$$\left(\frac{f}{g}\right)'(x_0) = \frac{f'(x_0)g(x_0) - f(x_0)g'(x_0)}{g(x_0)^2}.$$

T Derivative of the Inverse:

• Let f be bijective and differentiable with $f'(x_0) \neq 0$. Then

$$(f^{-1})'(f(x_0)) = \frac{1}{f'(x_0)}.$$

T Necessary Condition for Extremum:

• If x_0 is a local extremum and f is differentiable at x_0 , then

$$f'(x_0) = 0.$$

T Characterization of Local Extrema:

• If x_0 is a local extremum, then at least one holds:

- x_0 is a boundary point

• f is not differentiable at x_0

• $f'(x_0) = 0$

T Rolle's Theorem:

• Let f be continuous on $[a, b]$ and differentiable on (a, b) with $f(a) = f(b)$. Then there exists $\xi \in (a, b)$ such that

$$f'(\xi) = 0.$$

T Mean Value Theorem:

• Let f be continuous on $[a, b]$ and differentiable on (a, b) . Then there exists $\xi \in (a, b)$ such that

$$f'(\xi) = \frac{f(b) - f(a)}{b - a}.$$

T l'Hospital's Rule (0/0):

• If

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x) = 0$$

and

$$\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} = L,$$

then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = L.$$

T l'Hospital's Rule (∞/∞):

• If

$$\lim_{x \rightarrow a} |f(x)| = \lim_{x \rightarrow a} |g(x)| = \infty$$

and

$$\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} = L,$$

then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = L.$$

T l'Hospital at Infinity:

• If

$$\lim_{x \rightarrow \infty} f(x), g(x) \in \{0, \infty\}$$

and

$$\lim_{x \rightarrow \infty} \frac{f'(x)}{g'(x)} = L,$$

then

$$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = L.$$

T Monotonicity via Derivative:

•

$$f'(x) \geq 0 \iff f \text{ is increasing}$$

C Constant Functions:

• f is constant iff

$$f'(x) = 0 \quad \forall x.$$

T Convexity via First Derivative:

- f is convex iff f' is increasing.

C Convexity via Second Derivative:

- If f is twice differentiable:

$$f''(x) \geq 0 \iff f \text{ is convex.}$$

$$f''(x) \leq 0 \iff f \text{ is concave}$$

T Second Derivative Test:

- Let f be twice differentiable near x_0 and suppose $f'(x_0) = 0$.

Then: $f''(x_0) > 0 \Rightarrow x_0$ is a local minimum,

and $f''(x_0) < 0 \Rightarrow x_0$ is a local maximum.

5.2 Taylor Approximation and Power Series**Local Approximation by Polynomials:**

- In a sufficiently small neighborhood of a point x_0 , the graph of a differentiable function f behaves almost like its tangent at x_0 . Hence, the function can locally be approximated by its tangent.

D Linear Approximation / Tangent Approximation:

- If f is differentiable at x_0 , then

$$f(x) \approx f(x_0) + f'(x_0)(x - x_0).$$

The expression

$$P_1(x) = f(x_0) + f'(x_0)(x - x_0)$$

is called the *linearization*, *tangent approximation*, or *first Taylor polynomial* of f at x_0 .

T Taylor's Theorem:

- Let f be infinitely differentiable on an open interval I containing x_0 . Then for every $x \in I$,

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \cdots + \frac{1}{n!} f^{(n)}(x_0)(x - x_0)^n,$$

for some c between x_0 and x . Equivalently,

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k + R_n(x),$$

where

$$R_n(x) = \frac{1}{(n+1)!} f^{(n+1)}(c)(x - x_0)^{n+1}.$$

The polynomial

$$P_n(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k$$

is called the n -th *Taylor polynomial*.

Taylor Approximation and Power Series:

- Taylor polynomials provide increasingly accurate approximations of

many functions. For many functions, taking infinitely many terms recovers the original function.

D Taylor Series:

- For a function f , the series

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!} (x - a)^k$$

is called the *Taylor series* (or Taylor expansion) of f around the point a .

T Important Taylor Series:

- The following series converge for all x :

$$\sin(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$$

$$\cos(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots$$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

T Further Important Power Series:

- The following series converge for $-1 < x < 1$.

$$\ln(1+x) = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^n}{n} = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots$$

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \cdots$$

$$(1+x)^p = \sum_{n=0}^{\infty} \binom{p}{n} x^n = 1 + px + \frac{p(p-1)}{2!} x^2 + \cdots$$

Operations on Taylor Series:

- New Taylor series can be constructed from known ones using:
 - substitution
 - multiplication
 - differentiation
 - integration

T Multiplication of Power Series:

- Let

$$f(x) = \sum_{n=0}^{\infty} a_n x^n, \quad g(x) = \sum_{n=0}^{\infty} b_n x^n,$$

with x inside both convergence intervals. Then

$$f(x)g(x) = \sum_{n=0}^{\infty} \left(\sum_{k=0}^n a_k b_{n-k} \right) x^n.$$

T Termwise Differentiation:

- Let

$$f(x) = \sum_{n=0}^{\infty} a_n x^n$$

inside its convergence interval. Then f is differentiable and

$$f'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1}.$$

T Termwise Integration:

- Let

$$f(x) = \sum_{n=0}^{\infty} a_n x^n$$

inside its convergence interval I . Then for every interval $[a, b] \subseteq I$,

$$\int_a^b f(x) dx = \sum_{n=0}^{\infty} \int_a^b a_n x^n dx.$$

6 Differential equations

D Differential Equation:

- A *differential equation* is an equation involving an unknown function together with its derivatives.

$$u'(t) = r(a - u(t))(b - u(t))$$

where r, a, b are constants, another example

$$y^{(4)}(x) - 4y^{(3)}(x) + 5y''(x) - 4y'(x) + 4y(x) = 0.$$

The variables t, x are called *independent variables*, while u, y are called *dependent variables*.

D Linear Differential Equation:

- A differential equation of the form

$$a_n(x)y^{(n)}(x) + a_{n-1}(x)y^{(n-1)}(x) + \cdots + a_1(x)y'(x) + a_0(x)y(x) = s(x)$$

is called a *linear differential equation* with coefficients $a_i(x)$.

The *order* of the differential equation is the highest order derivative appearing in the equation. The function $s(x)$ is called the *forcing term*. If $s(x) = 0$, the equation is called *homogeneous*; otherwise it is called *inhomogeneous*.

D Differential Equation with Constant Coefficients:

- If all coefficient functions $a_i(x)$ are constant functions, the equation is called a *differential equation with constant coefficients*.

T Superposition Principle:

- If $y_1(x)$ and $y_2(x)$ are solutions of a linear homogeneous differential equation, then

$$y(x) = C_1 y_1(x) + C_2 y_2(x), \quad C_1, C_2 \in \mathbb{R},$$

is also a solution of the same differential equation.

T Fundamental Solutions:

- A linear homogeneous differential equation of order n possesses n solutions $y_1(x), \dots, y_n(x)$ such that the general solution is given by

$$y(x) = C_1 y_1(x) + \cdots + C_n y_n(x), \quad C_1, \dots, C_n \in \mathbb{R}.$$

These solutions are called *fundamental solutions*.

Characteristic Polynomial:

- Consider the differential equation

$$a_n u^{(n)}(x) + a_{n-1} u^{(n-1)}(x) + \cdots + a_1 u'(x) + a_0 u(x) = 0.$$

Using the ansatz $u(x) = e^{\lambda x}$ (Euler ansatz), we obtain the *characteristic equation*

$$a_n \lambda^n + a_{n-1} \lambda^{n-1} + \cdots + a_1 \lambda + a_0 = 0.$$

We get the *characteristic polynomial*:

$$p(\lambda) = a_n \lambda^n + a_{n-1} \lambda^{n-1} + \cdots + a_1 \lambda + a_0$$

T Solution of Linear Homogeneous Differential Equations:

- Consider the differential equation

$$a_n u^{(n)}(x) + a_{n-1} u^{(n-1)}(x) + \cdots + a_1 u'(x) + a_0 u(x) = 0, \quad a_i \in \mathbb{R}.$$

If all roots λ_i of the characteristic polynomial are real and have multiplicities m_i , then $u(x) = \sum_{i=1}^r \left(\sum_{p=0}^{m_i-1} C_{ip} x^p e^{\lambda_i x} \right)$.

If there are additionally complex conjugate roots $a_j \pm ib_j$ with multiplicities m_j , then $u(x) = \sum_{i=1}^r \left(\sum_{p=0}^{m_i-1} C_{ip} x^p e^{\lambda_i x} \right) + \sum_{j=1}^s \left(\sum_{q=0}^{m_j-1} A_{jq} x^q e^{a_j x} \sin(b_j x) + B_{jq} x^q e^{a_j x} \cos(b_j x) \right)$.

Constants in the General Solution:

- The general solution of a differential equation of order n contains n constants. Therefore, in order to determine a unique solution, one must specify n additional conditions.

D Boundary Value Problem:

- Conditions imposed at different points, e.g. for $n = 2$:

$$u(t_0) = u_0, \quad u(t_1) = u_1, \quad t_0 \neq t_1,$$

define a *boundary value problem*.

D Initial Value Problem:

- Conditions imposed at a single point, e.g. for $n = 2$:

$$u(t_0) = u_0, \quad u'(t_0) = v_0,$$

define an *initial value problem*.

7 Integration

8 Integration

D Antiderivative / Indefinite Integral:

- Let $I \subseteq \mathbb{R}$ be an interval and let

$$f : I \rightarrow \mathbb{R}$$

be a function. A differentiable function

$$F : I \rightarrow \mathbb{R}$$

satisfying

$$F'(x) = f(x) \quad \forall x \in I$$

is called an *antiderivative* of f on I . Determining F from f is called *indefinite integration*.

Non-Uniqueness of Antiderivatives:

- An antiderivative is not unique. If $F(x)$ is an antiderivative of $f(x)$, then

$$F(x) + C$$

is also an antiderivative for every constant $C \in \mathbb{R}$. Hence a function possesses an entire family of antiderivatives. This family is called the *indefinite integral* and is written as

$$\int f(x) dx.$$

The function f is called the *integrand*.

T Linearity of the Indefinite Integral:

- For indefinite integrals:

$$\int (f(x) \pm g(x)) dx = \int f(x) dx \pm \int g(x) dx$$

and

$$\int s f(x) dx = s \int f(x) dx,$$

where s is a constant.

T Basic Integration Rules:

- Rules:

- $\int k dx = kx + C$
- $\int x^n dx = \frac{1}{n+1} x^{n+1} + C, \quad n \neq -1$
- $\int e^x dx = e^x + C$
- $\int \frac{1}{x} dx = \ln(|x|) + C$
- $\int \sin(x) dx = -\cos(x) + C$
- $\int \cos(x) dx = \sin(x) + C$

- $\int \sinh(x) dx = \cosh(x) + C$

- $\int \cosh(x) dx = \sinh(x) + C$

Differentiation Rules vs. Integration Rules:

- Recall the two fundamental differentiation rules:

$$\frac{d}{dx} f(g(x)) = f'(g(x)) \cdot g'(x)$$

(chain rule) and

$$\frac{d}{dx} (f(x)g(x)) = f'(x)g(x) + f(x)g'(x)$$

(product rule). The goal is to derive and apply the corresponding rules in integration theory.

T Integration by Parts:

- Using the product rule:

$$\frac{d}{dx} (f(x)g(x)) = f'(x)g(x) + f(x)g'(x),$$

we obtain

$$\int \frac{d}{dx} (f(x)g(x)) dx = (fg)(x).$$

Therefore,

$$(fg)(x) = \int f'(x)g(x) dx + \int f(x)g'(x) dx.$$

Rearranging gives the formula for *integration by parts*:

$$\int f'(x)g(x) dx = f(x)g(x) - \int f(x)g'(x) dx$$

T Integration by Substitution:

- Using the chain rule:

$$\frac{d}{dx} f(g(x)) = f'(g(x))g'(x),$$

we obtain

$$\int f'(g(x))g'(x) dx = f(g(x)) + C.$$

With the substitution

$$y = g(x),$$

this becomes

$$\int f'(g(x))g'(x) dx = \int \frac{df}{dy} dy.$$

Hence the substitution rule is

$$\int f'(g(x))g'(x) dx = f(g(x)) + C$$

9 Integration Part 2

Motivation:

- Given a velocity function $v(t)$ describing one-dimensional motion, natural questions are:

- What is the total distance traveled?
- What is the change in position?

D Partition of an Interval:

- Let $[a, b] \subseteq \mathbb{R}$ be a compact interval. A *partition* of $[a, b]$ is a finite set of points

$$a = x_0 < x_1 < \cdots < x_{n-1} < x_n = b.$$

The points x_i are called *subdivision points*. A partition

$$a = y_0 < y_1 < \cdots < y_m = b$$

is called a *refinement* of

$$a = x_0 < x_1 < \cdots < x_n = b$$

if

$$\{x_0, \dots, x_n\} \subseteq \{y_0, \dots, y_m\}.$$

D Step Function:

- A function

$$f : [a, b] \rightarrow \mathbb{R}$$

is called a *step function* if there exists a partition

$$a = x_0 < x_1 < \cdots < x_n = b$$

such that for every $k = 1, \dots, n$, the restriction

$$f|_{(x_{k-1}, x_k)}$$

is constant.

D Integral of a Step Function:

- Let

$$f : [a, b] \rightarrow \mathbb{R}$$

be a step function with respect to the partition

$$a = x_0 < x_1 < \cdots < x_n = b.$$

Then the integral of f over $[a, b]$ is defined by

$$\int_a^b f(x) dx := \sum_{k=1}^n c_k (x_k - x_{k-1}),$$

where c_k denotes the constant value of f on (x_{k-1}, x_k) .

D Upper and Lower Sums:

- Let

$$f : [a, b] \rightarrow \mathbb{R}.$$

The set of upper sums is defined by

$$U(f) := \left\{ \int_a^b s(x) dx \mid s \in TF, s \geq f \right\},$$

and the set of lower sums by

$$L(f) := \left\{ \int_a^b s(x) dx \mid s \in TF, s \leq f \right\},$$

where TF denotes the set of step functions.

D Riemann Integral:

- A bounded function

$$f : [a, b] \rightarrow \mathbb{R}$$

is called *Riemann integrable* if

$$\sup L(f) = \inf U(f).$$

In this case, the common value is called the *Riemann integral* and is denoted by

$$\int_a^b f(x) dx := \sup L(f) = \inf U(f).$$

The numbers a, b are called the lower and upper limits of integration, and f is called the integrand.

T Properties of the Riemann Integral:

- (i) (Linearity) If $f, g : [a, b] \rightarrow \mathbb{R}$ are integrable and $\alpha, \beta \in \mathbb{R}$, then

$$\int_a^b (\alpha f + \beta g)(x) dx = \alpha \int_a^b f(x) dx + \beta \int_a^b g(x) dx.$$

- (ii) (Monotonicity) If

$$f(x) \leq g(x)$$

for all x , then

$$\int_a^b f(x) dx \leq \int_a^b g(x) dx.$$

- (iii) (Triangle Inequality)

$$\left| \int_a^b f(x) dx \right| \leq \int_a^b |f(x)| dx.$$

- (iv) (Reversal of Integration Limits)

$$\int_a^b f(x) dx = - \int_b^a f(x) dx.$$

- (v) (Splitting the Interval) For $a \leq c \leq b$:

$$\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx.$$

T Monotone Functions are Integrable:

- Every monotone function

$$f : [a, b] \rightarrow \mathbb{R}$$

is Riemann integrable.

T Continuous Functions are Integrable:

- Every continuous function

$$f : [a, b] \rightarrow \mathbb{R}$$

is Riemann integrable.

D Pointwise Convergence of Functions:

- Let

$$(f_n)_{n \in \mathbb{N}_0}$$

be a sequence of functions

$$f_n : D \rightarrow \mathbb{R},$$

and let

$$f : D \rightarrow \mathbb{R}.$$

The sequence (f_n) converges *pointwise* to f if for every $x \in D$, the sequence

$$(f_n(x))_{n \in \mathbb{N}_0}$$

converges to $f(x)$.

D Uniform Convergence:

- The sequence (f_n) converges *uniformly* to f if

$$\forall \varepsilon > 0 \exists N$$

such that for all

$$n \geq N \quad \text{and all } x \in D,$$

we have

$$|f_n(x) - f(x)| < \varepsilon.$$

T Uniform Limit of Continuous Functions:

- Let

$$(f_n)_{n \in \mathbb{N}_0}$$

be a sequence of continuous functions

$$f_n : D \rightarrow \mathbb{R}$$

converging uniformly to

$$f : D \rightarrow \mathbb{R}.$$

Then f is continuous.

T Uniform Limit and Integration:

- Let

$$(f_n)_{n \in \mathbb{N}_0}$$

be a sequence of integrable functions

$$f_n : [a, b] \rightarrow \mathbb{R}$$

converging uniformly to

$$f : [a, b] \rightarrow \mathbb{R}.$$

Then f is integrable and

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \int_a^b f_n(x) dx.$$

T Mean Value Theorem for Integrals:

- If

$$f : [a, b] \rightarrow \mathbb{R}$$

is continuous, then there exists

$$c \in [a, b]$$

such that

$$f(c) = \frac{1}{b-a} \int_a^b f(x) dx.$$

The quantity

$$\frac{1}{b-a} \int_a^b f(x) dx$$

is called the *average value* of f on $[a, b]$.

T Fundamental Theorem of Calculus:

- Let

$$f : [a, b] \rightarrow \mathbb{R}$$

be continuous. Then for every constant $C \in \mathbb{R}$, the function

$$F(x) := \int_a^x f(y) dy + C$$

is an antiderivative of f . Moreover, every antiderivative of f is of this form.

C Integral Representation:

- Let

$$F : [a, b] \rightarrow \mathbb{R}$$

be continuously differentiable. Then

$$F(x) = F(a) + \int_a^x F'(t) dt.$$

C Newton–Leibniz Formula:

- Let

$$f : [a, b] \rightarrow \mathbb{R}$$

be continuous and let F be an antiderivative of f . Then

$$\int_a^b f(t) dt = F(b) - F(a).$$

T Integration by Parts:

- Let

$$f, g : [a, b] \rightarrow \mathbb{R}$$

be continuously differentiable. Then

$$\int_a^b f(x)g'(x) \, dx = f(x)g(x) \Big|_a^b - \int_a^b f'(x)g(x) \, dx.$$

T Substitution Rule (Version 1):

- Let

$$f : I \rightarrow J$$

be continuously differentiable and

$$g : J \rightarrow \mathbb{R}$$

continuous. Then for every

$$[a, b] \subseteq I,$$

$$\int_a^b g(f(x))f'(x) \, dx = \int_{f(a)}^{f(b)} g(y) \, dy.$$

T Substitution Rule (Version 2):

- Let

$$f : I \rightarrow J$$

be continuously differentiable and

$$g : J \rightarrow \mathbb{R}$$

continuous. Assume

$$f'(x) \neq 0 \quad \forall x \in [a, b]$$

and let

$$f^{-1} : [f(a), f(b)] \rightarrow \mathbb{R}$$

denote the inverse of $f|_{[a,b]}$. Then

$$\int_a^b g(f(x)) \, dx = \int_{f(a)}^{f(b)} g(y)(f^{-1})'(y) \, dy.$$

10 Additional

Important Limits

- $\lim_{x \rightarrow \infty} \frac{1}{x} = 0$
- $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right) = 1$
- $\lim_{x \rightarrow 0} \frac{\log(1-x)}{x} = -1$
- $\lim_{x \rightarrow 0} x \log x = 0$
- $\lim_{x \rightarrow \infty} e^x = \infty$
- $\lim_{x \rightarrow -\infty} e^x = 0$
- $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}$
- $\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1$
- $\lim_{x \rightarrow \infty} e^{-x} = 0$
- $\lim_{x \rightarrow -\infty} e^{-x} = \infty$
- $\lim_{x \rightarrow 0} \frac{x}{\arctan x} = 1$
- $\lim_{x \rightarrow \infty} \arctan x = \frac{\pi}{2}$
- $\lim_{x \rightarrow \infty} \frac{e^x}{x^m} = \infty$
- $\lim_{x \rightarrow -\infty} x e^x = 0$
- $\lim_{x \rightarrow \infty} \left(\frac{x}{x+k}\right)^x = e^{-k}$
- $\lim_{x \rightarrow \infty} \ln x = \infty$
- $\lim_{x \rightarrow 0} \ln x = -\infty$
- $\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \ln(a)$
- $\lim_{x \rightarrow 0} \frac{e^{ax} - 1}{x} = a$
- $\lim_{x \rightarrow \infty} (1+x)^{1/x} = 1$
- $\lim_{x \rightarrow 0} (1+x)^{1/x} = e$
- $\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} = 1$
- $\lim_{x \rightarrow 1} \frac{\ln x}{x-1} = 1$
- $\lim_{x \rightarrow \infty} \frac{\ln x}{x} = 0$
- $\lim_{x \rightarrow \infty} \frac{\log x}{x^\alpha} = 0$
- $\lim_{x \rightarrow \infty} x^a q^x = 0, \ 0 \leq q < 1$
- $\lim_{x \rightarrow \infty} n^{1/n} = 1$
- $\lim_{x \rightarrow \infty} \sqrt[n]{x} = 1$
- $\lim_{x \rightarrow \infty} \frac{2x}{2^x} = 0$
- $\lim_{x \rightarrow \pm \infty} \left(1 + \frac{1}{x}\right)^x = e$
- $\lim_{x \rightarrow \infty} \left(1 - \frac{1}{x}\right)^x = \frac{1}{e}$
- $\lim_{x \rightarrow \pm \infty} \left(1 + \frac{k}{x}\right)^{mx} = e^{km}$
- $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$
- $\lim_{x \rightarrow \infty} \frac{\sin x}{x} = 0$
- $\lim_{x \rightarrow 0^+} x \ln x = 0$
- $\lim_{x \rightarrow (\pi/2)^-} \tan x = +\infty$
- $\lim_{x \rightarrow (\pi/2)^+} \tan x = -\infty$
- $\lim_{x \rightarrow 0} \frac{1}{\cos x} = 1$
- $\lim_{x \rightarrow 0} \frac{\cos x - 1}{x} = 0$

Rules of derivation (holds only if f^{-1} exists and f continuous.)

- Sum:** $(f + g)' = f' + g'$
- Product:** $(fg)' = f'g + fg'$
- Chain rule:** $(g \circ f)' = g'(f)f'$
- Quotient rule:** $\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$
- Derivative of the Inverse:** $(f^{-1})'(y_0) = \frac{1}{f'(x_0)}, \quad y_0 = f(x_0)$

Rules of integration

- Linearity of Integration:** $\int (f + g) = \int f + \int g$ and $\int cf = c \int f$
- Integration by Parts:** $\int f'g = fg - \int fg'$
- Substitution:** $\int g(f(x))f'(x)dx = \int g(y)dy$

Logarithms

- Change of base:** $\log_a x = \frac{\ln x}{\ln a}$
- Power rule:** $\log_a(x^y) = y \log_a x$
- Product rule:** $\log_a(xy) = \log_a x + \log_a y$
- Quotient rule:** $\log_a\left(\frac{x}{y}\right) = \log_a x - \log_a y$
- Logarithm 1:** $\log_a(1) = 0 \ \forall a \in \mathbb{N}$ (negative undefined)

Trigonometry

- Tangent and cotangent:** $\tan x = \frac{\sin x}{\cos x}, \quad \cot x = \frac{\cos x}{\sin x}$
- Pythagorean identity:** $\sin^2 x + \cos^2 x = 1$
- Double-angle sine:** $\sin(2x) = 2 \sin x \cos x$
- Double-angle cosine:** $\cos(2x) = \cos^2 x - \sin^2 x$
- Angle addition sine:** $\sin(x + y) = \sin x \cos y + \cos x \sin y$
- Angle addition cosine:** $\cos(x + y) = \cos x \cos y - \sin x \sin y$
- Even/Odd symmetry:** $\cos(x) = \cos(-x)$ and $\sin(-x) = -\sin(x)$
- Reflection at $\pi/2$:** $\cos(\pi - x) = -\cos(x)$ and $\sin(\pi - x) = \sin(x)$

Special Trigonometric Values

°	rad	$\sin(\xi)$	$\cos(\xi)$	$\tan(\xi)$
0°	0	0	1	0
30°	$\frac{\pi}{6}$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{3}}{3}$
45°	$\frac{\pi}{4}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	1
60°	$\frac{\pi}{3}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$
90°	$\frac{\pi}{2}$	1	0	$\emptyset$
120°	$\frac{2\pi}{3}$	$\frac{\sqrt{3}}{2}$	$-\frac{1}{2}$	$-\sqrt{3}$
135°	$\frac{3\pi}{4}$	$\frac{\sqrt{2}}{2}$	$-\frac{\sqrt{2}}{2}$	-1
150°	$\frac{5\pi}{6}$	$\frac{1}{2}$	$-\frac{\sqrt{3}}{2}$	$-\frac{\sqrt{3}}{3}$
180°	π	0	-1	0

Hyperbolic

- Hyperbolic sine:** $\sinh(x) := \frac{e^x - e^{-x}}{2} : \mathbb{R} \rightarrow \mathbb{R}$
- Hyperbolic cosine:** $\cosh(x) := \frac{e^x + e^{-x}}{2} : \mathbb{R} \rightarrow [1, \infty)$
- Hyperbolic tangent:** $\tanh(x) := \frac{\sinh(x)}{\cosh(x)} = \frac{e^x - e^{-x}}{e^x + e^{-x}} : \mathbb{R} \rightarrow (-1, 1)$
- Inverse hyperbolic sine:** $\operatorname{arsinh} x = \ln(x + \sqrt{x^2 + 1})$
- Inverse hyperbolic cosine:** $\operatorname{arcosh} x = \ln(x + \sqrt{x^2 - 1})$
- Inverse hyperbolic tangent:** $\operatorname{artanh} x = \frac{1}{2} \ln \frac{1+x}{1-x}$

Complement trick

$$\sqrt{ax+b} - \sqrt{cx+d} = \frac{ax+b-(cx+d)}{\sqrt{ax+b} + \sqrt{cx+d}}$$

Integration by parts

Should we get unavoidable cycle, where we have to integrate the same thing again, we may simply add the integral to both sides, and we thus have 2 times the integral on the left side and then finish the integration by parts on the right hand side and in the end divide by the factor up front to get the result.

Derivatives and Antiderivatives

Antiderivative	Function	Derivative
$\frac{x^{n+1}}{n+1}$	x^n	nx^{n-1}
$\ln x $	$\frac{1}{x} = x^{-1}$	$-x^{-2} = -\frac{1}{x^2}$
$\frac{2}{3}x^{3/2}$	$\sqrt{x} = x^{1/2}$	$\frac{1}{2\sqrt{x}}$
$\frac{n}{n+1}x^{1/n+1}$	$\sqrt[n]{x} = x^{1/n}$	$\frac{1}{n}x^{1/n-1}$
e^x	e^x	e^x
$\exp(x)$	$\exp(x)$	$\exp(x)$
$\frac{1}{a(n+1)}(ax+b)^{n+1}$	$(ax+b)^n$	$n(ax+b)^{n-1}a$
$x(\ln x -1)$	$\ln(x)$	$\frac{1}{x}$
$\frac{1}{\ln(a)}a^x$	a^x	$a^x \ln(a)$
$\frac{x}{\ln(a)}(\ln x -1)$	$\log_a x $	$\frac{1}{x \ln(a)}$
$-\cos(x)$	$\sin(x)$	$\cos(x)$
$\sin(x)$	$\cos(x)$	$-\sin(x)$
$-\ln \cos(x) $	$\tan(x)$	$\frac{1}{\cos^2(x)}$
$\ln \sin(x) $	$\cot(x)$	$-\frac{1}{\sin^2(x)}$
$x \arcsin(x) + \sqrt{1-x^2}$	$\arcsin(x)$	$\frac{1}{\sqrt{1-x^2}}$
$x \arccos(x) - \sqrt{1-x^2}$	$\arccos(x)$	$-\frac{1}{\sqrt{1-x^2}}$
$x \arctan(x) - \frac{1}{2} \ln(x^2+1)$	$\arctan(x)$	$\frac{1}{x^2+1}$
$\ln \sin(x) $	$\cot(x)$	$-\frac{1}{\sin^2(x)}$
$\cosh(x)$	$\sinh(x)$	$\cosh(x)$
$\sinh(x)$	$\cosh(x)$	$\sinh(x)$
$\ln \cosh(x) $	$\tanh(x)$	$\frac{1}{\cosh^2(x)}$

$F(x)$ (further derivatives $\Rightarrow$)	$f(x) = F'(x)$
$\frac{1}{a} \ln ax+b $	$\frac{1}{ax+b}$
$\frac{ax}{c} - \frac{ad-bc}{c^2} \ln cx+d $	$\frac{a(cx+d)-c(ax+b)}{(cx+d)^2}$
$\frac{x}{2} \sqrt{a^2+x^2} + \frac{a^2}{2} \ln x+\sqrt{a^2+x^2} $	$\sqrt{a^2+x^2}$
$\frac{x}{2} \sqrt{a^2-x^2} + \frac{a^2}{2} \arcsin\left(\frac{x}{ a }\right)$	$\sqrt{a^2-x^2}$
$\frac{x}{2} \sqrt{x^2-a^2} - \frac{a^2}{2} \ln x+\sqrt{x^2-a^2} $	$\sqrt{x^2-a^2}$
$\ln x+\sqrt{x^2 \pm a^2} $	$\frac{1}{\sqrt{x^2 \pm a^2}}$
$\arcsin\left(\frac{x}{ a }\right)$	$\frac{1}{\sqrt{a^2-x^2}}$
$\frac{1}{a} \arctan\left(\frac{x}{a}\right)$	$\frac{1}{a^2+x^2}$
$-\frac{1}{a} \cos(ax+b)$	$\sin(ax+b)$
$\frac{1}{a} \sin(ax+b)$	$\cos(ax+b)$
x^x	$x^x(1+\ln x)$
$(x^x)^x$	$(x^x)^x(x+2x\ln x)$
x^{x^x}	$x^{x^x}(x^{x-1}+x^x(1+\ln x)\ln x)$
$\frac{1}{2}(x-\frac{1}{2}\sin(2x))$	$\sin^2(x)$
$\frac{1}{2}(x+\frac{1}{2}\sin(2x))$	$\cos^2(x)$
$\operatorname{arsinh}(x)$	$\frac{1}{\sqrt{1+x^2}}$
$\operatorname{arcosh}(x)$	$\frac{1}{\sqrt{x^2-1}}$
$\operatorname{artanh}(x)$	$\frac{1}{1-x^2}$